



Made in USA

**TRUNNION MOUNTED
BALL VALVE
INSTALLATION, OPERATION,
MAINTENANCE MANUAL**

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WARNING

BEFORE ATTEMPTING TO REPAIR A VALVE THAT HAS BEEN IN SERVICE, BLOCK THE ENDS AND SLOWLY OPEN AND CLOSE THE VALVE TO VENT ANY INTERNAL PRESSURE.

BALON TRUNNION MOUNTED BALL VALVE STORAGE

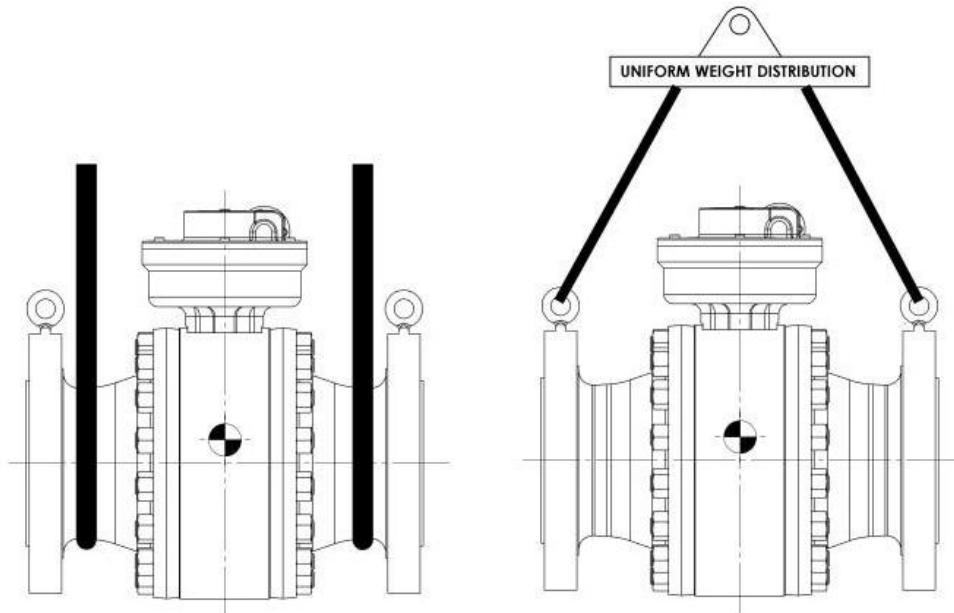
Balon Corporation ships ball valves in the open position. In order to protect valves during shipment, Balon ships trunnion mounted valves in wooden boxes. The hand wheels on gear-operated valves are removed and secured to the valves for shipment.

Valves should always be stored in the fully open position and should be kept indoors under clean, dry conditions.

BALON TRUNNION MOUNTED BALL VALVE INSTALLATION

Balon ball valves seal bi-directionally and may be installed in either direction.

Valves should be installed in the open position. Flange end protectors should be left in place while valves are being handled and should not be removed until final installation. Heavy valves should be lifted into position using suitable straps. The straps should be positioned around the conduits near the flanges or through the eye bolts, if provided. The straps should never be positioned around a gear operator, handwheel, or actuator.



Industry accepted make-up practice should be followed when installing flanged end valves.

Trash, such as welding slag, dirt, sand, and other foreign matter, should always be flushed from the line before operating any valve.

Piping systems are often pressure tested prior to being placed in service. Assure that all valves are in the open position during the introduction of the test medium. With valves in the open position or partially open position, valve shell testing may be done at a pressure up to 1.5 times the rated working pressure of the valves. Valves should be kept in the open position or partially open position during shell testing. Valve shell testing should never be done with the valves in the closed position.

With valves in the fully closed position, valve seat testing may be done at pressures up to 1.1 times the rated working pressure of the valves.

With all valves in the fully open position to avoid the entrapment of debris, the pressure test medium should be completely eliminated from the system after testing has been completed. Water that is allowed to remain trapped inside valves can freeze. Water that freezes inside valves can damage the valves and any pipe, fittings, or equipment directly or indirectly connected to the valve.

BALON TRUNNION MOUNTED BALL VALVE OPERATION

A ball valve is designed to be used in the fully open or fully closed position. On gear operated valves, the position indicator is located on the top of the gear box. To open the valve, rotate the handwheel counterclockwise. To close the valve, rotate the handwheel clockwise. Using a ball valve in the partially open position can result in seat damage and, under certain conditions, damage to other valve components or to pipe, fittings, or equipment directly or indirectly connected to the valve. Applying excessive torque in an attempt to turn a ball beyond its 90 degree stop point could damage the valve stops and misalign the ball.

A valve should always be operated within the pressure and temperature limits designated by the manufacturer. See Table 1 and Table 2 for limits.

Table 1: WCB / A105 Class 1 Pressure (psi) and Temperature (F°) Limits				
Pressure Class	150	300	600	900
-20° - 100°	285	740	1480	2220
250°	245	668	1335	2000

Table 2: LCC / LF2 Class 1 Pressure (psi) and Temperature (F°) Limits				
Pressure Class	150	300	600	900
-50° - 100°	285	740	1480	2220
250°	245	668	1335	2000

It is good practice to operate any valve periodically in order to assure that the ball is able to move freely.

The Balon trunnion mounted ball valve is equipped with a body bleed fitting that can be used for draining the body cavity or for performing a double block and bleed procedure. Draining the body can be done with the valve in either the closed or the open position. The double block and bleed procedure must be performed with the valve in the closed position. Locate the bleed fitting and determine the position of the bleed hole. Be aware that fluid and debris will be vented under pressure through this hole.

⚠ WARNING

Position yourself and others safely out of line of venting pressure. A trunnion mounted ball valve can unexpectedly release residual pressure from the body cavity past the downstream seat. Valve users should exercise caution by protecting personnel and equipment from spontaneous pressure release.

Under certain circumstances it is possible for trash and debris to effectively plug the body bleed fitting. This can give the false impression that the body cavity is depressurized. Before attempting to dismantle any ball valve, the valve should be opened and closed a complete cycle to assure that there is no pressure trapped within the body cavity.

MAINTENANCE FOR BALON TRUNNION MOUNTED BALL VALVES

Balon trunnion mounted ball valves require no regular maintenance to maintain proper operation and sealing. Balon stems require no adjustment or greasing for proper sealing and operation.

SECONDARY SEALING IN BALON TRUNNION MOUNTED BALL VALVES

Balon trunnion mounted valves are equipped with grease injection fittings that may be used when attempting to restore the seat seal or to deliver grease to the surface of the ball. Grease should be injected into the upstream injection fitting when attempting to restore the seat seal. Always assure the compatibility of injected grease with the valve components and the medium in the line.

For cut sheets and pressure and temperature information, please see:

<https://www.balon.com/downloads/CompleteCatalog.pdf>

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Annex 1 PED Information

This is supplemental information to be used in addition to preceding information within the IOM.

Valves are considered to be Cat III to module H. Valves and nameplate are marked CE 0062.

Maximum Design Conditions are indicated on valve nameplate(s). These design conditions shall not be exceeded.

Valves are intended to be used for controlling service common to oil and gas operations. This includes, but not limited to, mixed hydrocarbons and produced water. PED group 1, fluids and/or gases. Do not use valves for any other purpose without first consulting the manufacturer.

Users of the valves are responsible for properly and safely installing valves in appropriate service applications.

Valves do not include safety relief devices. The user is responsible for providing a properly sized relieving device within piping systems to ensure piping system pressure remains below the valve's maximum allowable pressure.

Do not subject the valves to thermal or mechanical shock.

Manufacturer has not considered cyclic loading, external pressure, vibration, or decomposition of unstable fluids in the valve design. Manufacturer did not design the valves for use under external fire conditions.

To prevent operational failures, user must maintain the safe operation of the entire piping system within which the valve is installed.

Annex 1 cont. PED Information

Risk Assessment Numbers

The following information is a risk associated with the installation, maintenance, and decommissioning of valves that the Risk Assessment Number is higher than 11. Table 1

Risk	Risk Assessment Number (RAN)	Preventive Measure (PMN)
Some valves are equipped with lifting lugs. Improper lifting can cause damage to lifting lugs and hence be a potential cause for personal injury	11	3, 4, 5
Attention should be given to the environment in which any fitting, maintenance, and disassembly work is being done. Serious bodily injury, including death, may result if user fails to eliminate all sources of ignition/explosion.	15	1, 2, 3, 5, 6, 8, 9
Any fitting, maintenance, or disassembly should be performed with care.	11	1, 2, 3, 5, 6, 9, 10
Welding valves into piping systems generates heat. This can cause the valve to be hot enough to burn human skin, ignite flammable gases, or create other hazards.	15	1, 2, 3, 5, 6, 8, 9, 11
User must assess and ensure a suitable environment before installing, maintaining, or disassembling a valve.	15	1, 2, 5, 6, 7, 8, 9
User must exercise care when valves are fitted into piping systems with one end open to atmosphere and must not operate the valve from the closed to an open position if the system is under pressure.; otherwise, external leakage and / or serious injury, including death, could occur.	15	1, 2, 5, 9, 12, 13
Metallic piping systems can conduct electric current. Lightning strike, static charge, and any other sources of electric current should be considered as each is a potential source for ignition or explosion that could result in serious injury, including death.	15	1, 5, 8, 16
Ball valves must be used in the fully open or fully closed position. Never throttle through a ball valve because doing so could damage the valve or result in serious injury, including death.	11	1, 5, 6, 9, 17
User must exercise care to prevent unsafe exposure to service media. User must dispose properly of all debris or media removed from a valve and must dispose properly of the valve when it is decommissioned.	13	1, 5, 8, 9, 13, 14, 15

Annex 1 cont. PED Information

Preventive Measure

The following information outlines the preventive measures users must take to mitigate the risk. Table 2

PMN	Preventive Measure
1.	Use proper protective clothing and equipment
2.	Use proper tools
3.	Use caution and follow general safety guidelines.
4.	Use a sling or chain when lifting. Follow lifting manufacturer's guidelines when lifting. Obey all safety regulations/guidelines for lifting.
5.	Specific regulations for the equipment and standard safety regulations must be observed.
6.	See Installation, Operation, and Maintenance Manual (IOM)
7.	Balon valves are not suitable for oxygen service; a risk of combustion or explosion exists.
8.	Static charge can build-up causing an arc that could increase risk of combustion or explosion.
9.	The operator of the equipment is responsible for adhering to any and all applicable regulations.
10.	Use a high-quality thread compound when installing threaded end valves. A back-up wrench should be placed on the valve component (body or Adapter) nearest to the pipe connection that is being adjusted. Failure to do so may allow a valve joint to loosen or separate, which could cause serious injury, including death.
11.	User must follow use appropriate eye protection and follow proper welding procedures.
12.	Depressurize the piping system
13.	Provide a way to contain/capture system media.
14.	Maximum Allowable Working Pressure is displayed on the valve.
15.	Use caution before performing any work, fitting, maintenance, or disassembly
16.	Properly grounded piping system, use of insulation kits, & use of lightning protection are examples of ways to help protect against sources of ignition/explosion. Operator must adhere to all applicable regulations.
17.	Lock valve in the fully open or fully closed position.